

מבחן מועד ב

חלק א

תכנון

-- תכנון דקודר

;library IEEE

;use IEEE.STD_LOGIC_1164.ALL

entity Address_Decoder is

) Port

Address : in std_logic_vector(11 downto 0) -- כתובת כניסה בת 12 ביטים

Enable : out std_logic -- אות פלט של הפעלה

;

;end Address_Decoder

architecture Behavioral of Address_Decoder is

begin

process(Address)

begin

if (unsigned(Address) >= x"300" and unsigned(Address) <= x"310") then

Enable <= '1' -- הפלט יופעל אם הכתובת בטווח

else

Enable <= '0' -- אחרת הפלט יישאר כבוי

;end if

;end process

;end Behavioral

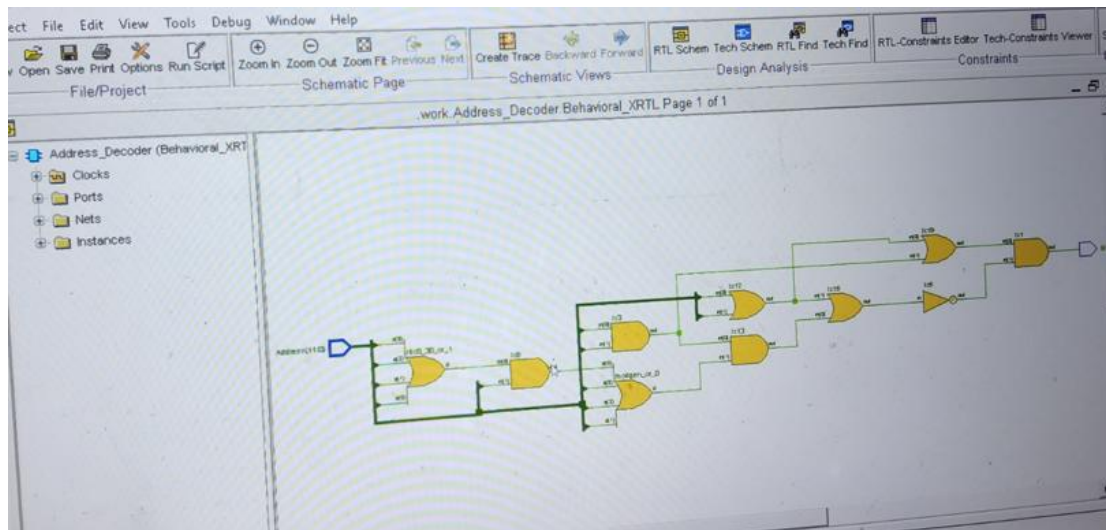
חלק ב

סינטזה

יוסי ברים
צוות 1

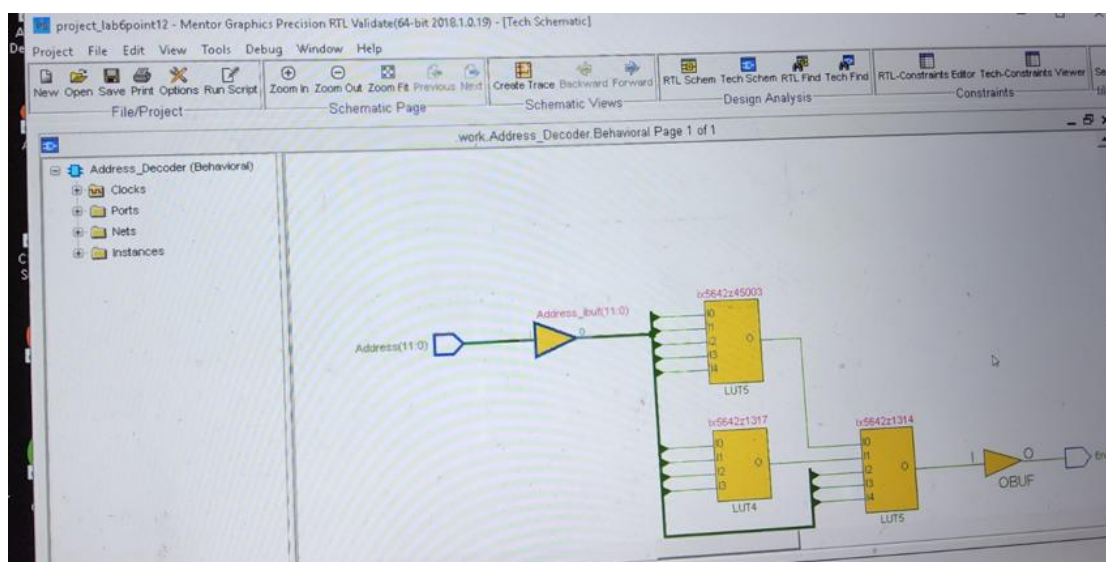
סעיף א

Rtl



סעיף ב

טכנולוגיה



סעיף ג

Area report

Precision RTL Validate 64-bit 2018.1.0.19 (Production Release) Fri Aug 10 05:55:53 PDT //
2018

//

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//

Running on Windows Server 2016 ybrim@MSTS-ELEC 10.0.17763 x64 //

//

Start time Sun Dec 15 17:44:40 2024 //

Device Utilization for 7A15TCPG236

Resource	Used	Avail	Utilization

IOs	13	106	12.26%
Global Buffers	0	32	0.00%
LUTs	3	10400	0.03%
CLB Slices	1	8150	0.01%
Dffs or Latches	0	20800	0.00%
Block RAMs	0	25	0.00%
DSP48E1s	0	45	0.00%

Library: work Cell: Address_Decoder View: Behavioral

Cell Library References Total Area

	IBUF	xca7	12 x	
	LUT4	xca7	1 x	1 LUTs
	LUT5	xca7	2 x	2 LUTs
	OBUF	xca7	1 x	

Number of ports :	13
Number of nets :	28
Number of instances :	16
Number of references to this view :	0

: Total accumulated area	
Number of LUTs :	3
Number of accumulated instances :	16

IO Register Mapping Report

Design: work.Address_Decoder.Behavioral

+-----+-----+-----+-----+-----+				
Port	Direction	INFF	OUTFF	TRIFF
+-----+-----+-----+-----+-----+				
			Address(11)	Input
+-----+-----+-----+-----+-----+				
			Address(10)	Input
+-----+-----+-----+-----+-----+				
			Address(9)	Input

[illegible]

Total registers mapped: 0

חלק ד

Timing report

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Running on Windows Server 2016 ybrim@MSTS-ELEC 10.0.17763 x64 //
//
Start time Sun Dec 15 17:44:40 2024 //

Device: Xilinx - ARTIX-7 : 7A15TCPG236 : 3 --

..CTE report summary --

POST-SYNTHESIS TIMING REPORTS ARE ESTIMATES AND SHOULD NOT BE RELIED ON TO --
MAKE QoR DECISIONS. For accurate timing information, please run place-and-route (P&R)
.and review P&R generated timing reports

=====

Clock Frequency Report

Domain	Clock Name	Min Period (Freq)	Required Period (Freq)
-----	-----	-----	-----
VirtualClock_PS	VirtualClock	4.740 (210.970 MHz)	10.000 (100.000 MHz)

=====

Setup Timing Analysis of VirtualClock

Setup Slack Path Summary

Data				Data		
Setup Path				Data	Start	End
Index	Slack	Delay	Source Clock	Dest. Clock	Pin	Pin
						Edge

VirtualClock	VirtualClock	Address(11)	Enable	Rise	4.740	5.260	1
VirtualClock	VirtualClock	Address(0)	Enable	Rise	4.740	5.260	2
VirtualClock	VirtualClock	Address(3)	Enable	Rise	4.740	5.260	3
VirtualClock	VirtualClock	Address(7)	Enable	Rise	4.740	5.260	4
VirtualClock	VirtualClock	Address(4)	Enable	Rise	4.740	5.260	5
VirtualClock	VirtualClock	Address(1)	Enable	Rise	4.740	5.260	6
VirtualClock	VirtualClock	Address(5)	Enable	Rise	4.740	5.260	7
VirtualClock	VirtualClock	Address(6)	Enable	Fall	4.740	5.260	8
VirtualClock	VirtualClock	Address(2)	Enable	Rise	4.740	5.260	9
VirtualClock	VirtualClock	Address(9)	Enable	Fall	4.348	5.652	10

CTE Path Report

:Critical path #1, (path slack = 5.260)

SOURCE CLOCK: name: VirtualClock period: 10.000000

Times are relative to the 1st rising edge

DEST CLOCK: name: VirtualClock period: 10.000000

Times are relative to the 2nd rising edge

NAME	GATE	DELAY	ARRIVAL	DIR	FANOUT
	Address(11)	(port)		0.000	dn
	Address(11)	(net)	0.000		1
	Address_ibuf(11)/I	IBUF		0.000	dn
	Address_ibuf(11)/O	IBUF	1.147	1.147	dn
	Address_int(11)	(net)	0.295		1
	ix5642z1317/I0	LUT4		1.442	dn
	ix5642z1317/O	LUT4	0.097	1.539	up
	nx5642z3	(net)	0.295		1

יוסי ברים
צוות 1

ix5642z1314/I1	LUT5		1.834	up
ix5642z1314/O	LUT5	0.097	1.931	up
nx5642z1	(net)	0.295		1
Enable_obuf/I	OBUF		2.226	up
Enable_obuf/O	OBUF	2.514	4.740	up
Enable	(net)	0.000		0
Enable	(port)		4.740	up

Initial edge separation: 10.000

Source clock delay: - 0.000

Dest clock delay: + 0.000

Edge separation: 10.000

Setup constraint: - 0.000

Data required time: 10.000

:Data arrival time

חלק ה

סינטזה

synthesize

.Info: [662]: Current working directory: Z:/vhdlab6/project_lab6point12_impl_1 #

Warning: [19539]: No debug point is implemented, continuing with the normal synthesis #
.run

Info: [15002]: Optimizing design <...>.Address_Decoder.Behavioral #

Info: [15002]: Optimizing design <...>.Address_Decoder.Behavioral #

...Info: [8010]: Gated clock transformations: Begin #

...Info: [8010]: Gated clock transformations: End #

Info: [1081]: Constraints: Precision will generate the following clock domain information #
against the clocks. Since the clock domain information is not provided by the user for the
below clocks, all clocks are assumed to be synchronous clocks. This will result in all paths
between the clock domains to be analyzed for timing. If this behaviour is not intended for

any clock domain or path, please override using -domain option or false path constraint.
Among these clocks, clock with the smallest period will be chosen as the default domain and
.the rest of the clock periods specified using multiply-by or divide-by factor

:Info: Constraints: Precision Generated Clock Domains #

Info: Domain Name: VirtualClock_PS Clock Name: VirtualClock #

.Info: [3027]: Writing file: <...>/project_lab6point12_impl_1/Address_Decoder.edf #

Info: -- Writing file Z:/vhdlab6/project_lab6point12_impl_1/Address_Decoder.tcl #

.Info: [3027]: Writing file: <...>/project_lab6point12_impl_1/Address_Decoder.xdc #

.Info: [3027]: Writing file: <...>/project_lab6point12_impl_1/Address_Decoder.v #

Info: -- Writing file Z:/vhdlab6/project_lab6point12_impl_1/Address_Decoder.tcl #

...Info: [12045]: Starting timing reports generation #

.Info: [12046]: Timing reports generation done #

Info: [12048]: POST-SYNTHESIS TIMING REPORTS ARE ESTIMATES AND SHOULD NOT BE #
RELIED ON TO MAKE QoR DECISIONS. For accurate timing information, please run place-and-
.route (P&R) and review P&R generated timing reports

.Info: [665]: Finished synthesizing design #

.Info: [11019]: Total CPU time for synthesis: 0.5 s secs #

.Info: [11020]: Overall running time for synthesis: 1.2 s secs #